

Sr No.	Unit No.	Question
1	1	What are the various categories of software? OR List different software application domains.
2	1	Name at least three characteristics of software. OR Explain characteristics of software.
3	1	Demonstrate your understanding of umbrella activities of a Software process.
4	1	if you have developed a word processing software product, what process model will you choose? justify the answer and explain.
5	1	Explain about the umbrella activities which support software development process and discuss about their necessity in maintaining the quality in both software process and product that is being developed for railway reservation system
6	1	Identify the umbrella activities in software engineering process. OR Discuss the importance of umbrella activities in software engineering.
7	1	Explain Software Engineering as a Layered technology OR Describe the layers of software engineering and their significance.
8	1	What is Process? Discuss the process framework activities.
9	1	Define software
10	1	What is Software Engineering? What is Process? What is Product?
11	1	Compare Product and Process
12	1	List two deficiencies in the waterfall model. Which model do you suggest to overcome each deficiency
13	1	Provide three examples of software projects that would be amenable to the waterfall model. Be specific
14	1	Summarize the pros and cons of the iterative software development model.
15	1	Compare Prototype and RAD Model. OR Compare and contrast the Waterfall Model and the Incremental Process Model.
16	1	Describe the various Evolutionary Process Model with neat diagram.
17	1	Discuss the prototype model. What is the effect of designing prototype on the overall cost of the software project?
18	1	What is the importance of Process Model in development of Software System? Explain Prototype Process Model.
19	1	Discuss about the various drawbacks of spiral model
20	1	Explain Spiral Model in brief with suitable diagram.
21	1	Explain the process model which is normally suits for development of large-scale software system.
22	1	Which is more important-the product or process? Justify your answer
23	1	Compare the life cycle models based on their distinguishing factors, strengths and weaknesses
24	1	Explain iterative waterfall and spiral model for software life cycle and discuss various activities in each phase.

25	1	Assume that you are the technical manager of a software development organization. A client approached you for a software solution the problems stated by the client have uncertainties which lead to loss if it not planned and solved which software development model you will suggest for this project. justify. Explain the model with its pros and cons and neat sketch
26	1	Define feasibility study.
27	2	Define agility in the context of software development.
28	2	List the values outlined in the Agile Manifesto.
29	2	Recall the principles of Agile software development.
30	2	Explain the concept of Extreme Programming (XP) and its key practices.
31	2	Define Scrum and its key components.
32	2	List and explain the roles within a Scrum team. OR Explain the significance of the Scrum Master role in a Scrum team. OR Explain Scrum Roles and Responsibilities of Product owner. OR Explain Scrum Roles and Responsibilities of Scrum Master. OR Explain Scrum Roles and Responsibilities of Development Team.
33	2	Recall the purpose of Scrum ceremonies such as Sprint Planning and Daily Standups. OR Explain term Daily Scrum Meeting.
34	2	Differentiate between Sprint Backlog and Product Backlog in Scrum.
35	2	Describe the principles of Adaptive Software Development (ASD) and how it differs from traditional approaches.
36	2	Differentiate between Scrum and other Agile methodologies.
37	2	Develop a user story for a Library Management software feature.
38	2	Create acceptance criteria for a user story.
39	2	Compare and contrast the burn-down chart and burn-up chart in Agile project management. OR Write a short note on burn-down chart.
40	2	Explain Scrum framework.
41	2	Explain Feature Driven Development (FDD).
42	2	Define Agile estimation and its importance in software development.
43	2	List and explain common Agile estimation techniques.
44	2	Estimate the effort required for implementing a set of user stories using Planning Poker.
45	2	Explain Planning Poker estimation technique.
46	2	Explain T-Shirt Sizes estimation technique.
47	2	List and explain different Scrum Phases.
48	2	List and explain different Sprint Phases. OR Write a short note on Product Backlog. OR Write a short note on Sprint Backlog.
49	3	Explain Requirement Engineering Tasks. OR

		Write a short note on Requirement Engineering. OR Write a short note on Requirement Engineering.
50	3	"Write functional and non-functional requirements of Hotel Management System OR List five functional requirement of Library management System."
51	3	Enlist characteristic of SRS. Write a SRS for Hospital management system
52	3	State the difference between procedural Design and Object-Oriented Design.
53	3	Explain Quality Function Deployment (QFD).
54	3	List and explain Characteristics of a Good SRS.
55	3	What is the work product of software design process and who does this?
56	3	How requirements are classified? list them with an example for each
57	3	What is activity diagram and Swim-lane? Draw activity diagram for book issue in library management.
58	3	Give a use case diagram for an online shopping which should provide provisions for registering authenticating the customers
59	3	Differentiate Procedural Design and Object Oriented Design
60	3	What is Architectural Design? Enlist different styles and patterns of architecture.
61	3	What is Object Oriented Design of a system? Draw the Use case diagram and Class diagram for Library Management system.
62	3	Describe Golden Rules of User Interface Design
63	3	Explain Design pyramid for WebApps
64	3	Compare Coupling and Cohesion. Explain different types of Coupling and its effects on software modules.
65	3	List out the various types of cohesion and coupling.
66	4	Explain various coding standard and guidelines.
67	4	List set of guidelines for BVA.Also Explain merits and demerits of BVA.
68	4	Explain White Box Testing With an Example
69	4	What are the different testing strategies? Explain any one with suitable example.
70	4	Explain concept of test case and scenario.
71	4	List and explain different approach for Code Review.
72	4	Write a short note on Software Documentation.
73	4	Define Test Scenarios and Test Case.
74	4	Difference between Verification & Validation.
75	4	Explain Unit testing.
76	4	Explain Regression testing. When to do regression testing?
77	4	Explain Smoke testing.
78	4	Explain Validation testing.
79	4	Explain System testing.
80	4	Explain Acceptance testing.
81	4	Explain Equivalence Partitioning.
82	4	What is Cyclomatic Complexity? Define Steps to find Cyclomatic Complexity using flow graph. OR

		What is Cyclomatic complexity? Define Steps to find Cyclomatic complexity using flow graph. OR Consider the program given below <pre>void main() { int i,j,k; readln (i,j,k); if ((i < j) (i > k)) { Writeln ("then part"); if (j < k) writeln ("j less then k"); else writeln (" j not less then k"); } else writeln("else Part"); }</pre> (i) Draw the flow graph. (ii) Determine the cyclomatic complexity. (iii) Arrive at all the independent paths.
83	4	Explain Integration testing.
84	4	Compare alpha and beta testing.
85	4	What are the different levels of testing? Briefly discuss the goal of each level.
86	4	Write a short note on Testing Web Applications.
87	4	Explain Software Quality Assurance (SQA).
88	4	Explain the significance of quality standards in software development. OR Explain Six Sigma quality standards. OR List the Six Sigma company. OR Explain CMM (Capability Maturity Model) Level.
89	4	Differentiate between software quality assurance and quality control.
90	4	Explain how software reliability is measured and evaluated.
91	4	Describe the process and objectives of Formal Technical Reviews (FTRs).
92	5	Discuss Software Project Management and W5HH Principle in brief.
93	5	Explain Software metrics used for software cost estimation.
94	5	Explain Process and Product Metrics.
95	5	Explain Size-Oriented Metrics.
96	5	Explain Function Oriented Metrics.
97	5	Explain Object-Oriented Metrics.
98	5	Explain Use Case Oriented Metrics.
99	5	Explain Function Point Metrics.
100	5	Explain Process & Problem Based Estimation.
101	5	Explain Empirical Estimation Models.
102	5	Explain COCOMO model for project estimation OR Write short notes on COCOMO model. OR

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		Explain COCOMO model for project estimation
103	5	Draw the Time-line chart for the Library Management System.
104	5	Draw the Time-line chart for the Hospital Management System.
105	5	Discuss RMMM. OR Explain RMMM OR Explain Risk Management OR What is Risk Management? Explain RMMM plan.
106	5	Explain Different type of Risk
107	5	Write a short note on Re-Engineering
108	5	Explain Reverse Engineering.
109	5	Explain Forward Engineering.
110	5	Explain Software Configuration Management.
111	5	Explain Types of Software Maintenance.
112	5	Define DevOps and its key principles.
113	5	List the benefits of adopting a component-based software engineering approach.
114	5	Explain Computer-Aided Software Engineering (CASE) tools.
115	5	Explain the relationship between DevOps, agile development, and continuous integration/continuous deployment (CI/CD).